5.12 Homework: Integration Review (calculator) — Markscheme

1. Plant growth regression. Data: (d, h) pairs for 7 days.

   (a) *GDC: Enter data into lists and run linear regression.*

   $a = 1.93258\ldots, \quad b = 7.21662\ldots$

   $a = 1.93, \quad b = 7.22$ **(A1)(A1)**

   (b) $r = 0.991087\ldots$

   $r = 0.991$ **(A1)**

   (c) Substitute $d = 20$ into the regression line: **(M1)**

   $h = 1.93(20) + 7.22 = 45.8$

   $h = 45.9$ cm (using unrounded values: $1.93258 \times 20 + 7.21662 = 45.868\ldots$) **(A1)**

   Total [5 marks]

2. Sam invests \$1700, nominal rate 2.74% compounded half-yearly.

   Note: All answers must be given correct to two decimal places.

   (a) *GDC: Finance solver with $N = 20$, $I\% = 2.74$, $PV = -1700$, $P/Y = 2$, $C/Y = 2$.*

   OR $1700\left(1 + \dfrac{0.0274}{2}\right)^{2\times10}$ **(M1)(A1)**

   $= \$2231.71$ **(A1)**

   (b) *GDC: Finance solver with $N = 10$, $PV = -1700$, $FV = 2231.71$, $P/Y = 1$, $C/Y = 1$.*

   OR $1700\left(1 + \dfrac{r}{100}\right)^{10} = 2231.71$ **(M1)**

   $r = 2.75876\ldots$

   $r = 2.76$ **(A1)**

   (c) Interest $= 2231.71 - 1700 = \$531.71$ **(A1)**

   Total [6 marks]

3. Geometric sequence: $u_1 = 50$, $u_4 = 86.4$.

   $50r^3 = 86.4$ **(A1)**

$r = \sqrt[3]{\dfrac{86.4}{50}} = 1.2$ **(A1)**

$S_n = \dfrac{50(1.2^n - 1)}{0.2} > 33\,500$ **(A1)**

GDC: Solve $250(1.2^n - 1) = 33500$ *or use table/graph.* **(M1)**

$n > 26.9045\ldots \quad (\text{or } S_{26} = 28368.8,\ S_{27} = 34092.6)$

$n = 27$ **(A1)**

Total [5 marks]

4. Two buildings with $PA = 150$, $PB = 90$, $AB = 154$.

(a) Cosine rule: **(M1)**

$154^2 = 150^2 + 90^2 - 2(150)(90)\cos(\angle APB)$ **(A1)**

$\cos(\angle APB) = \dfrac{150^2 + 90^2 - 154^2}{2(150)(90)} = \dfrac{22500 + 8100 - 23716}{27000} = \dfrac{6884}{27000}$

$\angle APB = 75.2286\ldots^\circ$

$\angle APB = 75.2^\circ$ (or 1.31 radians) **(A1)**

(b) Since both angles of elevation are θ: **(M1)**

$\theta = \dfrac{180^\circ - 75.2286\ldots^\circ}{2} = 52.3856\ldots^\circ$ **(M1)**

$h = 150\sin\theta = 150\sin(52.3856\ldots^\circ) = 118.820\ldots$

$h = 119$ m **(A1)**

Total [6 marks]

5. Ferris wheel: diameter 110 m, lowest point 10 m, period 20 min.

$h(t) = a\cos(bt) + c$

Amplitude $= \dfrac{110}{2} = 55$ **(A1)**

Since P starts at the lowest point: $a = -55$ **(A1)**

Centre height: $c = 10 + 55 = 65$ **(A1)**

$\dfrac{2\pi}{b} = 20$ **(M1)**

$b = \dfrac{\pi}{10}$ $(= 0.314\ldots)$ **(A1)**

Total [5 marks]

6. Drug decay: $A(t) = 500e^{-kt}$.

(a) $A(0) = 500$ mg **(A1)**

(b) $280 = 500e^{-3k}$ **(A1)**

$k = -\frac{1}{3}\ln\left(\frac{280}{500}\right) = 0.193272\ldots$

$k = 0.193$ **(A1)**

(c) $500e^{-0.193272\ldots T} = 140$ **(A1)**

$T = -\frac{1}{0.193272}\ln\left(\frac{140}{500}\right) = 6.58636\ldots$

$T = 6.59$ hours **(A1)**

Total [5 marks]

7. Carbon-14: $A = A_0e^{-kt}$, 100 units at death, half-life 5730 years.

(a) At $t = 0$: $A = A_0e^0 = A_0 = 100$ **(A1)(AG)**

(b) $50 = 100e^{-5730k}$ **(M1)**

$e^{-5730k} = \frac{1}{2}$

$-5730k = \ln\frac{1}{2} = -\ln 2$ **(A1)**

$k = \frac{\ln 2}{5730}$ **(A1)(AG)**

(c) 25% has decayed $\Rightarrow$ 75% remains, i.e. $A = 75$ **(A1)**

$75 = 100e^{-(\ln 2/5730)\,t}$ **(M1)**

$t = -\frac{5730}{\ln 2}\ln(0.75) = 2378.16\ldots$

$t = 2380$ years (to the nearest 10) **(A1)**

Total [7 marks]

8. Kinematics: $v = \dfrac{(t^2+1)\cos t}{4}, \quad 0 \le t \le 3.$

(a) Direction changes when $v = 0$: **(M1)**

$\cos t = 0 \Rightarrow t = \dfrac{\pi}{2} = 1.57079\ldots$

GDC: solve $v(t) = 0$ for $0 \leq t \leq 3$.

$t = 1.57$ s $\left(= \dfrac{\pi}{2}\right)$ **(A1)**

(b) Acceleration $a(t) = v'(t)$. **(M1)**

GDC: solve $v'(t) = -1.9$ for $0 \leq t \leq 3$.

$t_1 = 2.26277\ldots, \quad t_2 = 2.95736\ldots$

$t = 2.26$ s, $\quad t = 2.96$ s **(A1)(A1)**

(c) *GDC: The speed $|v|$ is greatest at $t = 3$.* **(A1)**

$a(3) = v'(3) = -1.83778\ldots$

$a = -1.84\ \mathrm{m\,s^{-2}}$ **(A1)**

Total [7 marks]

9. Kinematics: $v(t) = 4e^{-t/3}\cos\left(\dfrac{t}{2} - \dfrac{\pi}{4}\right), \quad 0 \leq t \leq 4\pi.$

(a) Acceleration is zero when $v'(t) = 0$. **(M1)**

GDC: solve $v'(t) = 0$.

$t_1 = 0.394791\ldots$

$t_1 = 0.395$ s $\left(= \arctan\dfrac{5}{12}\right)$ **(A1)**

(b) Particle at rest when $v(t) = 0$. **(M1)**

First zero: $t = \dfrac{3\pi}{2} = 4.71238\ldots$

Second zero: $t = \dfrac{7\pi}{2} = 10.9955\ldots$

$t_2 = 11.0$ s $\left(= \dfrac{7\pi}{2}\right)$ **(A1)**

(c) *GDC: Distance* $= \displaystyle\int_{t_1}^{t_2} |v(t)|\,\mathrm{d}t$ **(A1)**

$$= \int_{0.395}^{4.712} v\,\mathrm{d}t - \int_{4.712}^{10.996} v\,\mathrm{d}t = 6.538\ldots + 1.293\ldots$$

$= 7.83118\ldots$

Distance $= 7.83$ m **(A1)**

Total [6 marks]